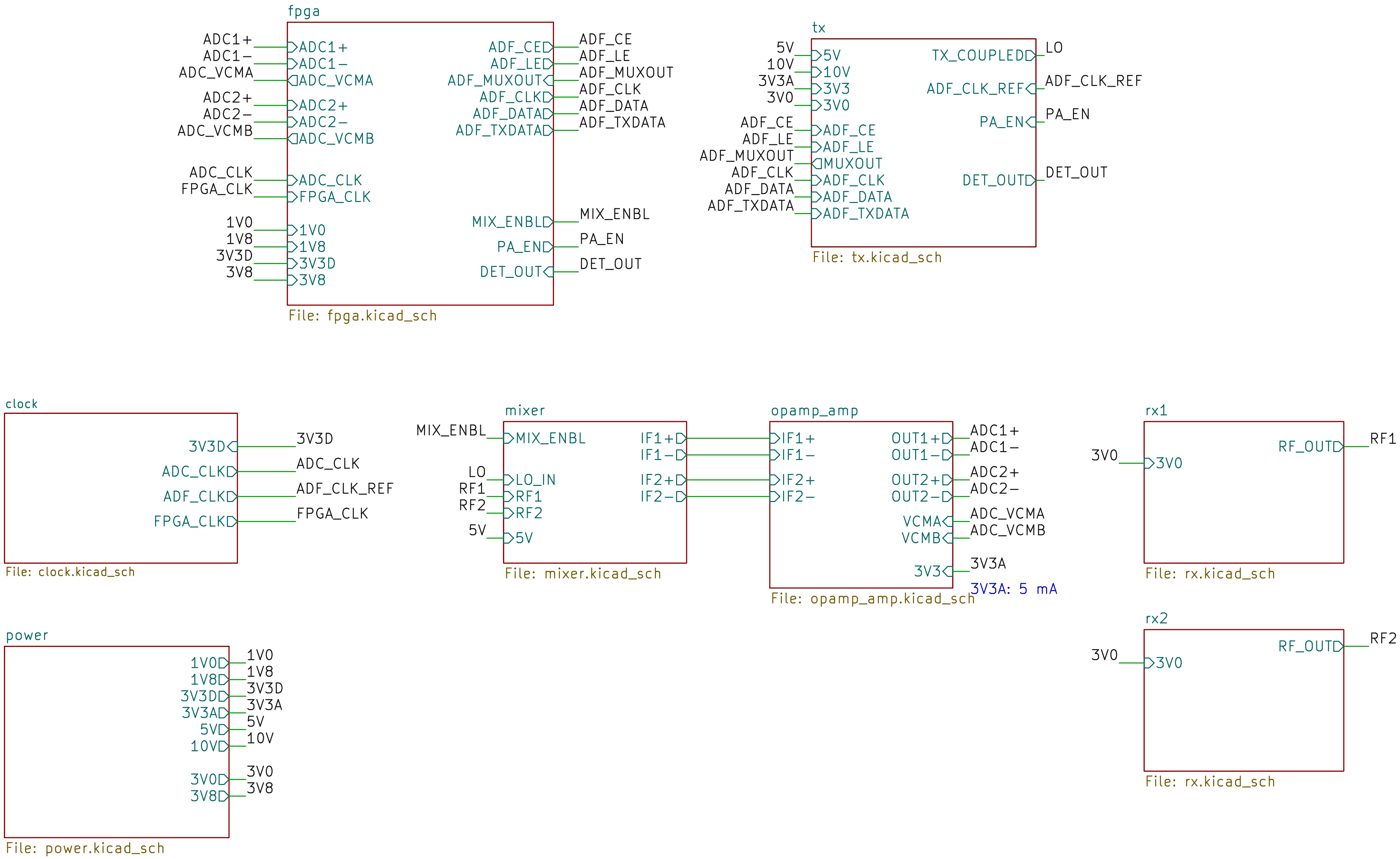
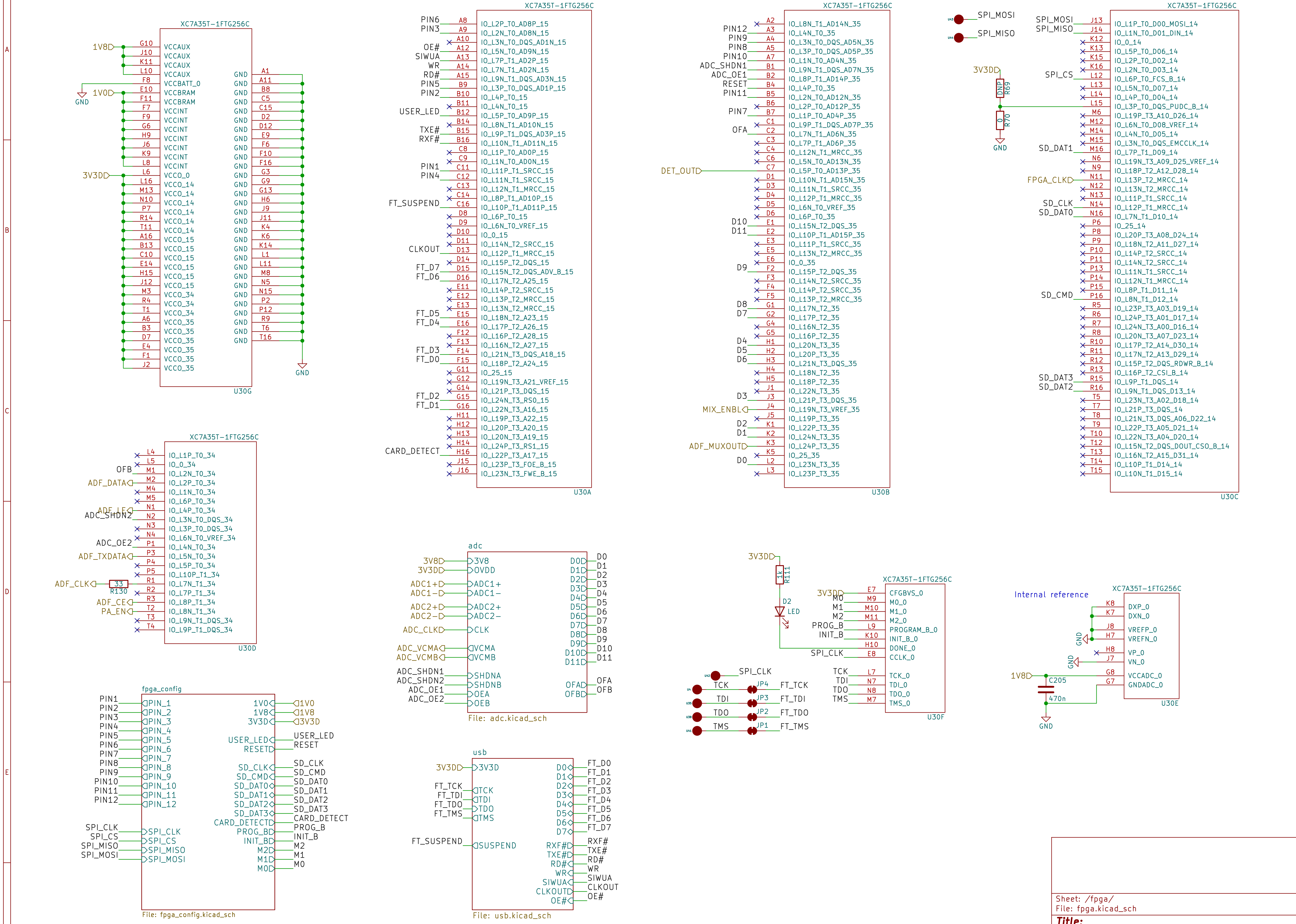


Sheet: /
File: fmcw3.kicad_sch

Title: FMCW3 by CENK KESKİN

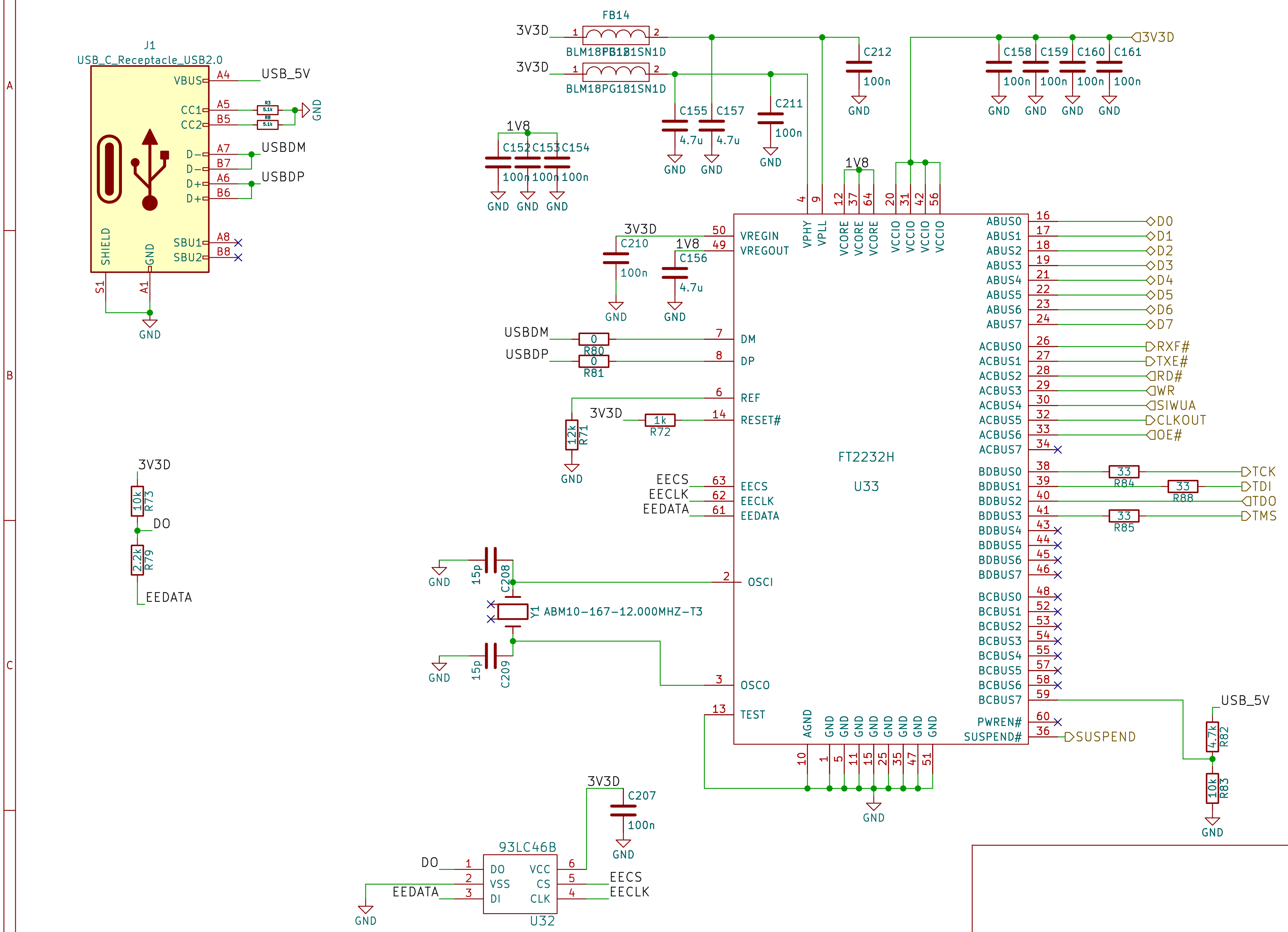
Size: A4 Date: Rev:
KiCad E.D.A. 9.0.7 Id: 1/12



Sheet: /fpga/
File: fpga.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 2/12



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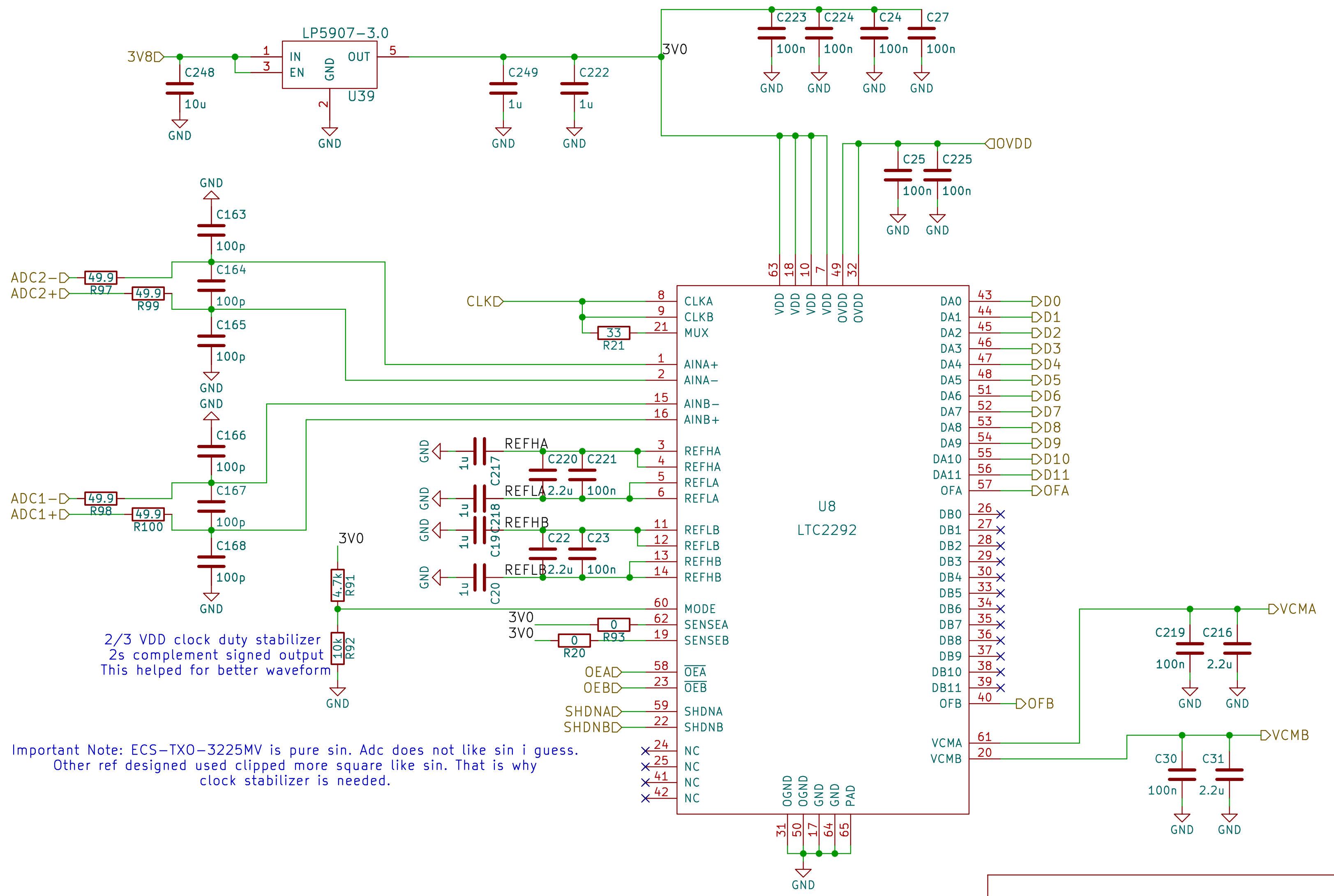
Size: A4 | Date: | Rev:
KiCad E.D.A. 9.0.7 | Id: 3/12



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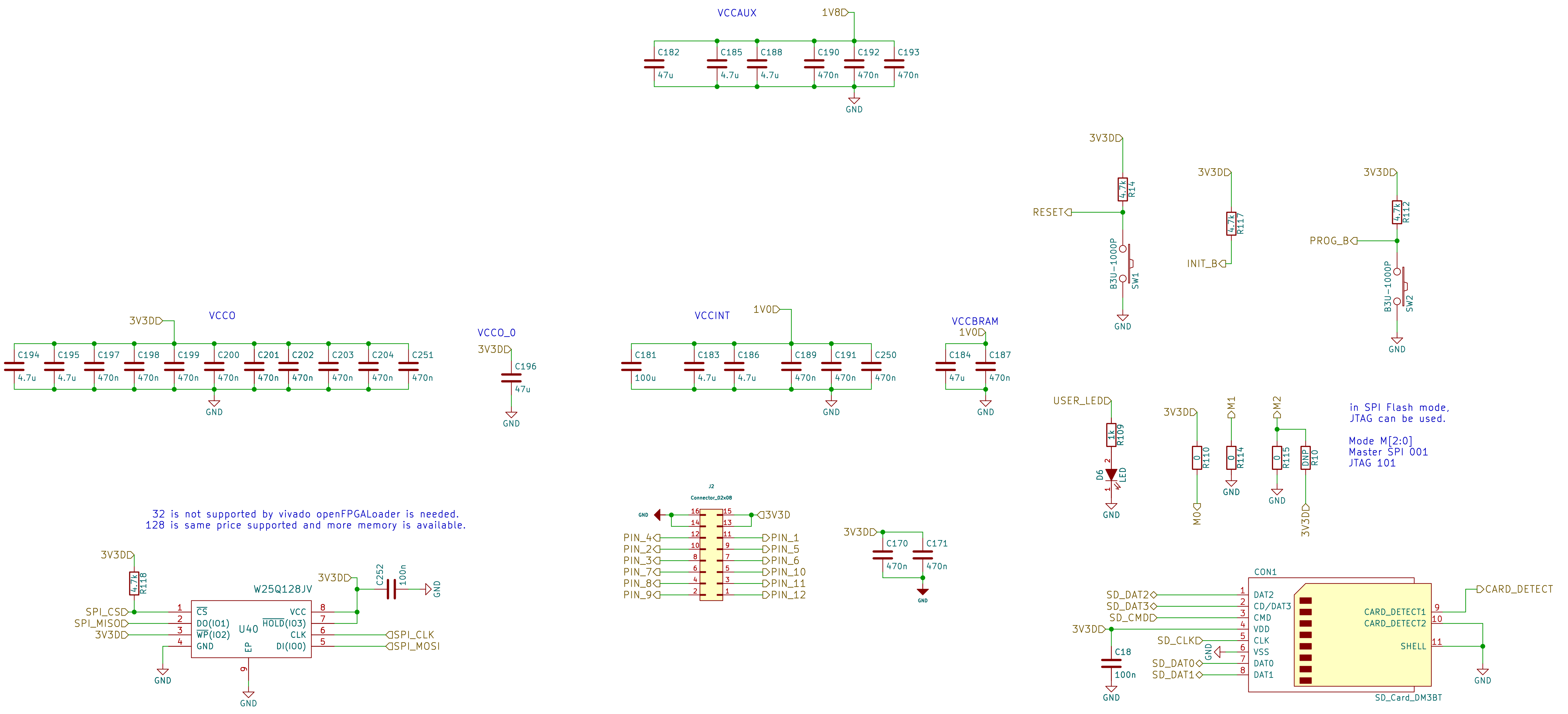
File: adc.kicad_sch

Title:

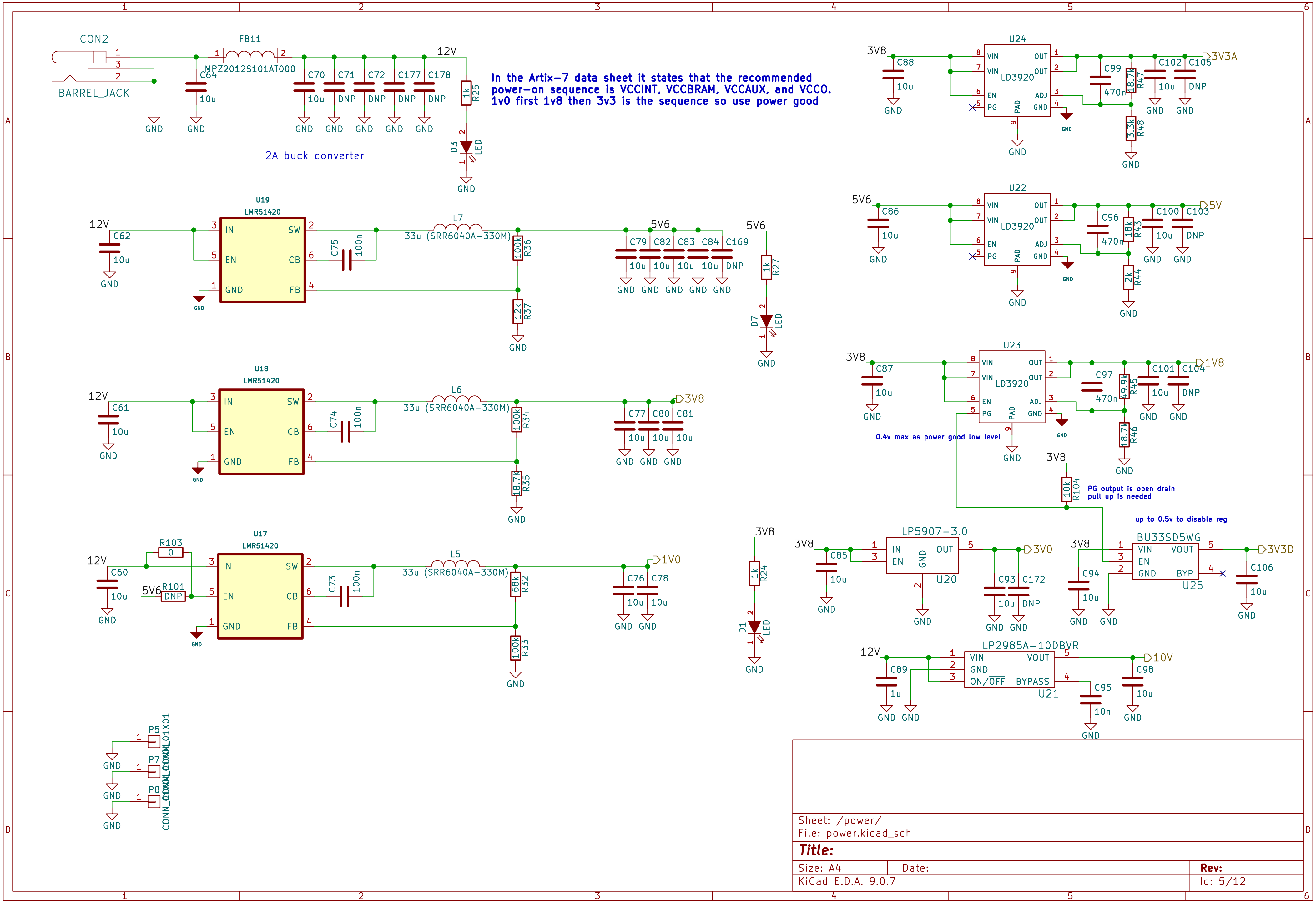
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 4/12

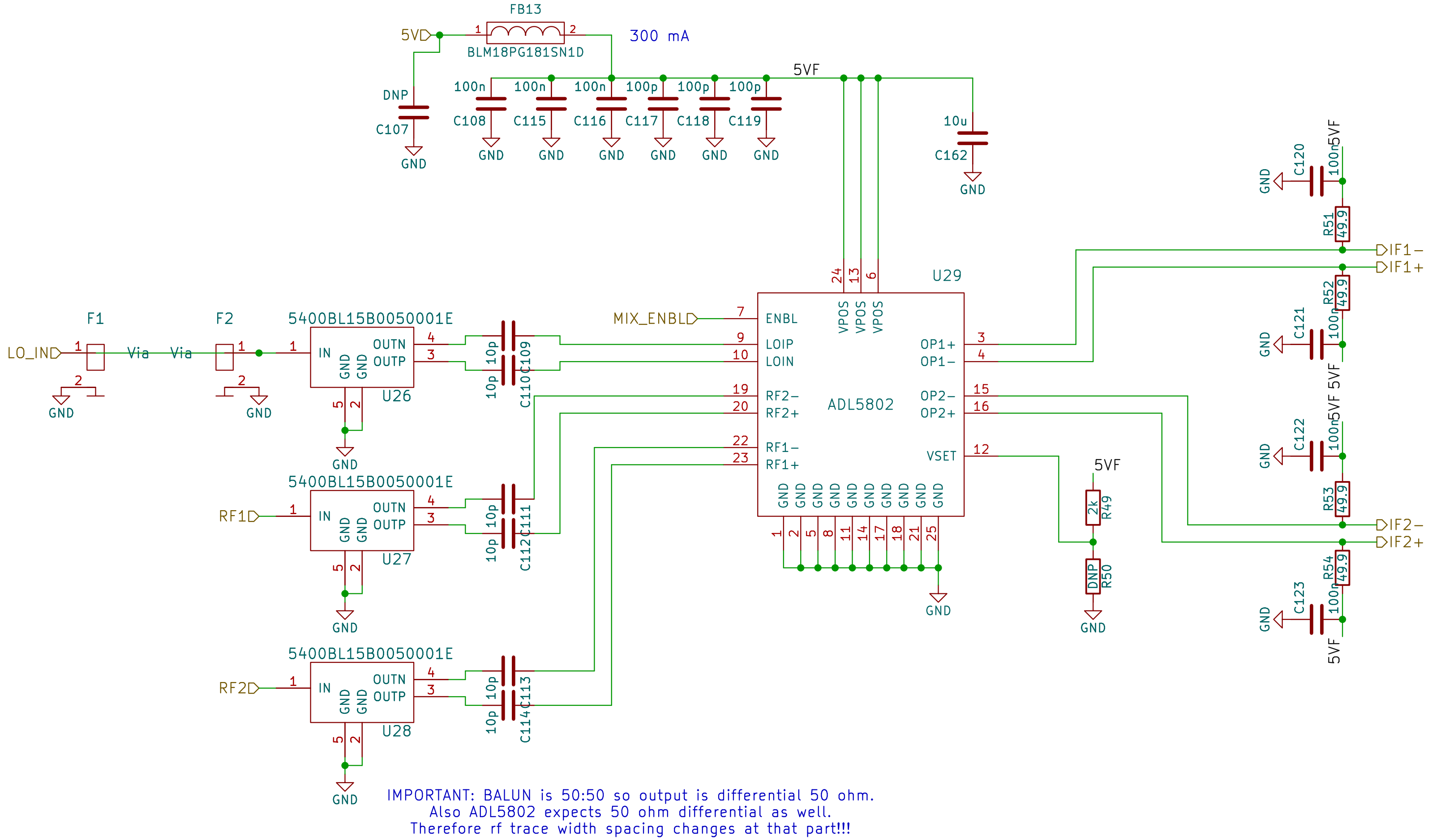


Sheet: /fpga/fpga_config/
File: fpga_config.kicad_sch

Title:

Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 12/12


In the Artix-7 data sheet it states that the recommended power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO. 1v0 first 1v8 then 3v3 is the sequence so use power good
2A buck converter
CON2
BARREL_JACK
FB11
MPZ2012S101AT000
C64 10u
C70 10u
C71 DNP
C72 DNP
C177 DNP
C178 DNP
12V
R25 1k
D3 LED
U19 LMR51420
U18 LMR51420
U17 LMR51420
IN SW EN CB GND FB
L7 33u (SRR6040A-330M)
L6 33u (SRR6040A-330M)
L5 33u (SRR6040A-330M)
C62 10u
C61 10u
C60 10u
C75 100n
C74 100n
C73 100n
R36 100k
R37 12k
R34 100k
R35 18.7k
R32 68k
R33 100k
R103 0
R101 DNP
5V6
3V8
1V0
C79 C82 C83 C84 C169
10u 10u 10u 10u DNP
C77 C80 C81
10u 10u 10u
C76 C78
10u 10u
R27 1k
D7 LED
R24 1k
D1 LED
U24 LD3920
U22 LD3920
U23 LD3920
VIN OUT EN ADJ PG GND PAD
3V3A
5V
1V8
C88 10u
C99 470n
R47 18.7k
R48 3.3k
C102 10u
C105 DNP
C86 10u
C96 470n
R43 18k
R44 2k
C100 10u
C103 DNP
C87 10u
C97 470n
R45 49.9k
R46 18.7k
C101 10u
C104 DNP
0.4v max as power good low level
R104 10k
PG output is open drain pull up is needed
up to 0.5v to disable reg
LP5907-3.0
U20
C85 10u
3V0
C93 10u
C172 DNP
BU33SD5WG
U25
VIN VOUT EN GND BYP
C94 10u
3V3D
C106 10u
LP2985A-10DBVR
U21
VIN VOUT GND ON/OFF BYPASS
C89 1u
C95 10n
C98 10u
10V
P5 P7 P8
CONN_01X01
GND
Sheet: /power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 5/12

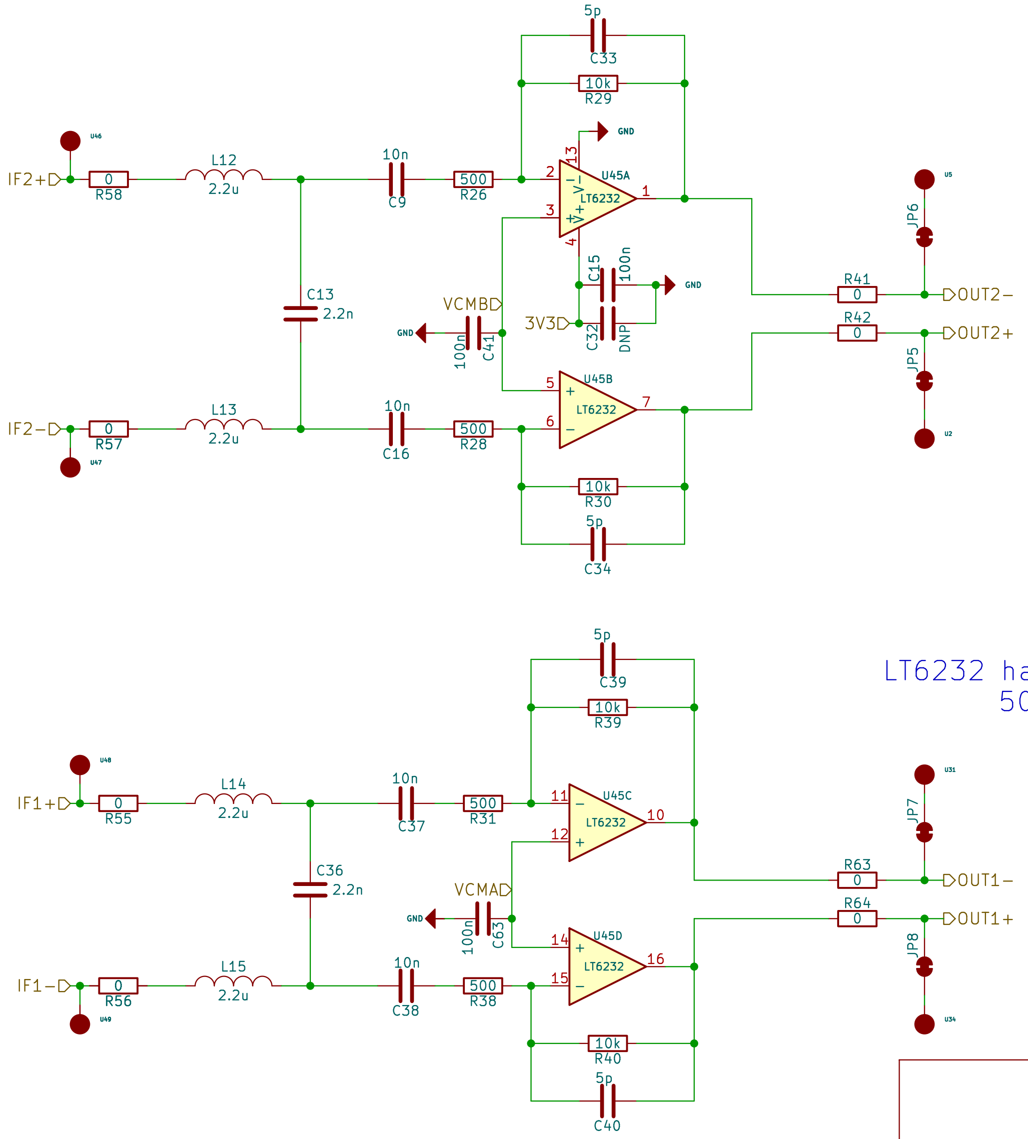


Sheet: /mixer/
File: mixer.kicad_sch

Title:

Size: A4 Date: **Rev:**

KiCad E.D.A. 9.0.7 Id: 6/12



FPGA FIR Filter works perfectly.
Second stage for HPF
range compensation filer is removed

LT6232 has better noise performance as 1.1nV/√Hz
500 10K has less noise contribution

Sheet: /opamp_amp/
File: opamp_amp.kicad_sch

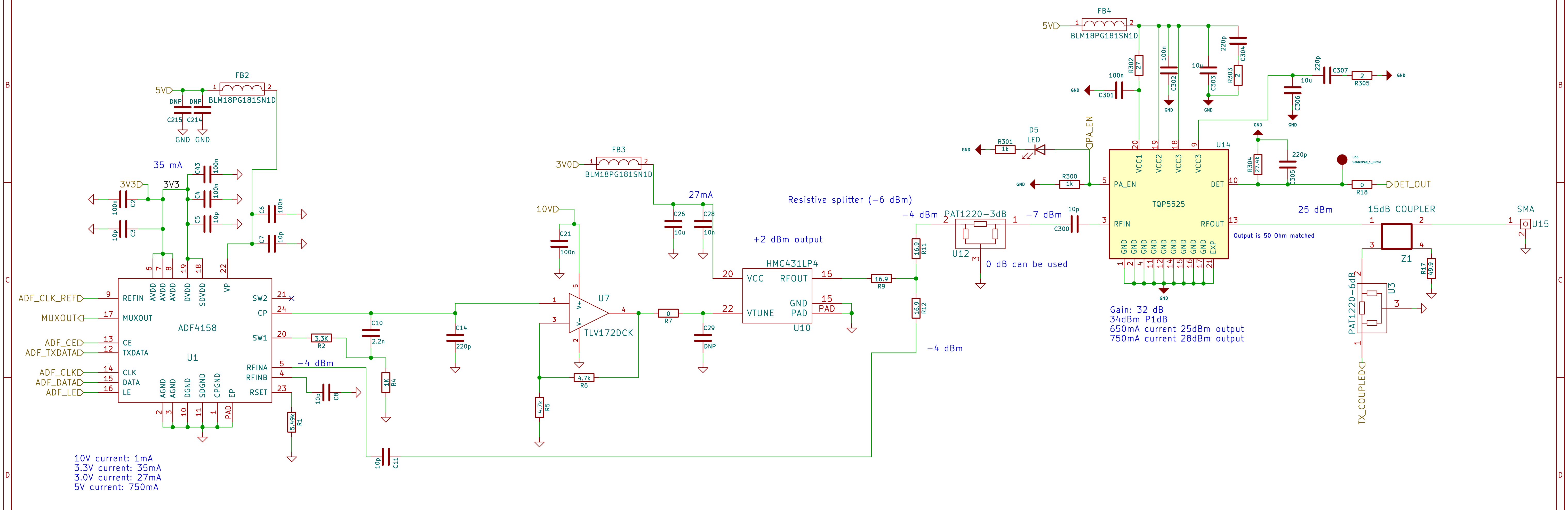
Title:

Size: A4 Date:

KiCad E.D.A. 9.0.7

Rev:

Id: 7/12

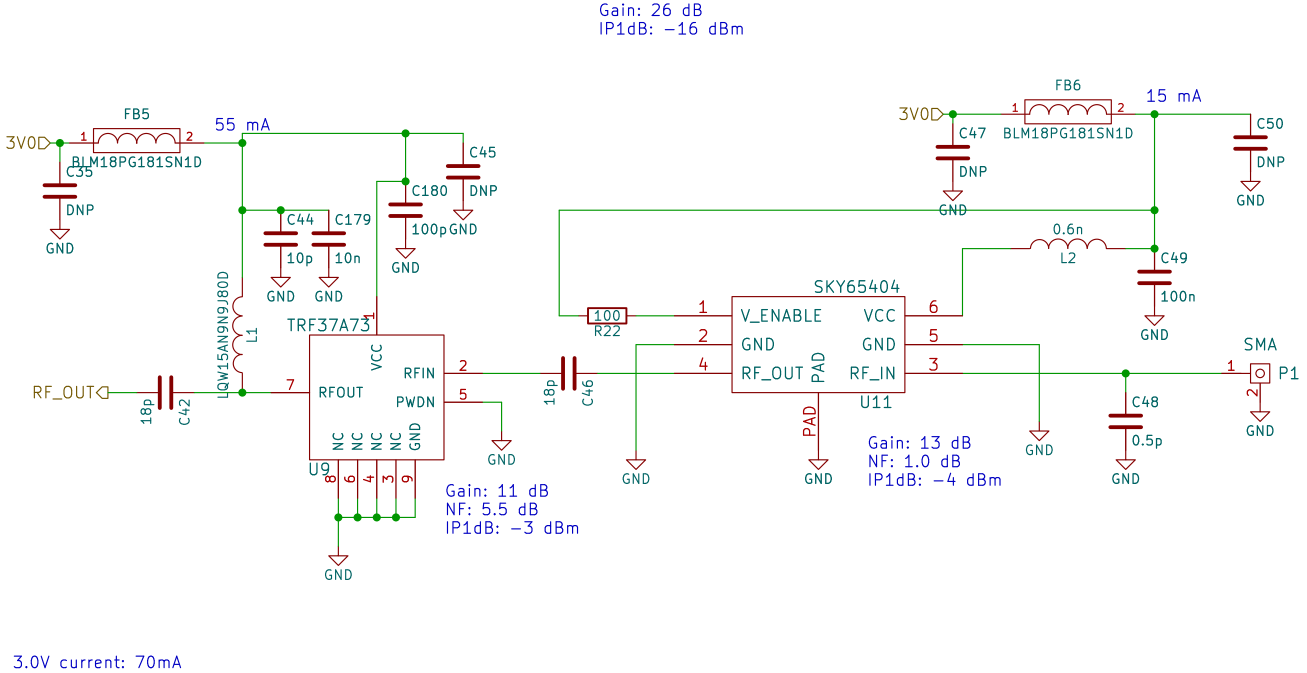


Sheet: /tx/
File: tx.kicad_sch

Title:

Size: A3 | Date: | Rev:

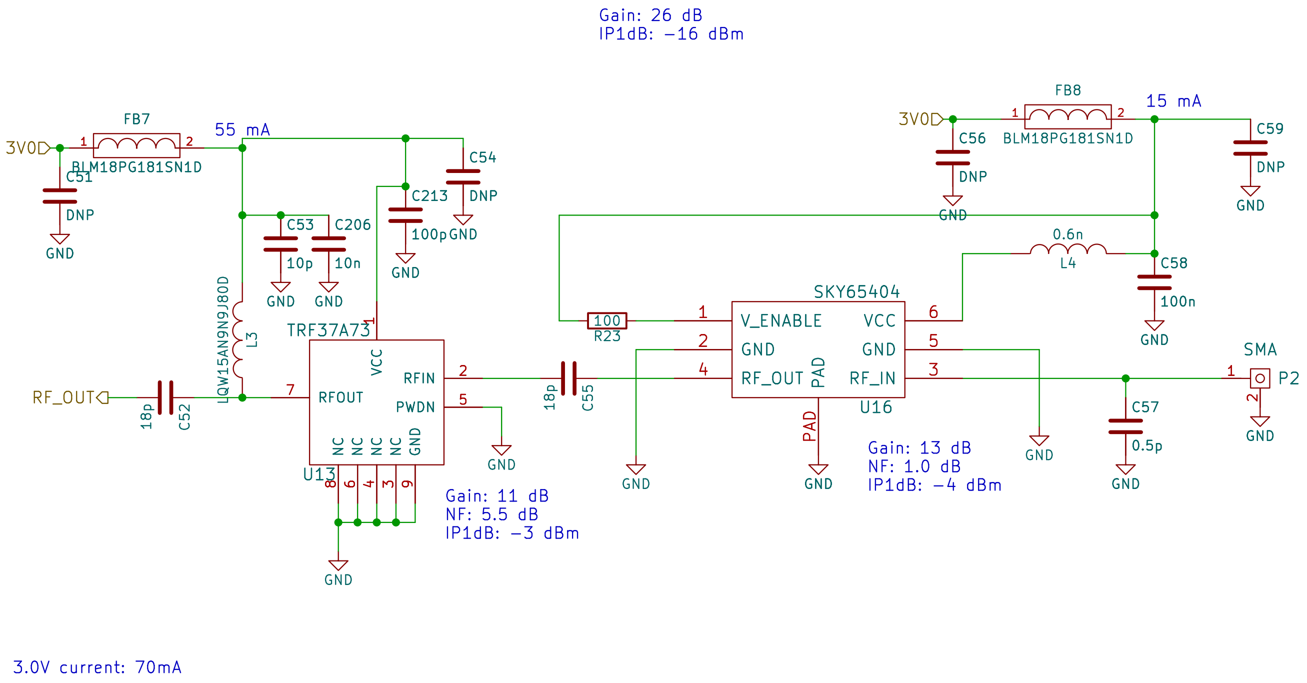
KiCad E.D.A. 9.0.7 | Id: 8/12



Sheet: /rx1/
File: rx.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 9/12

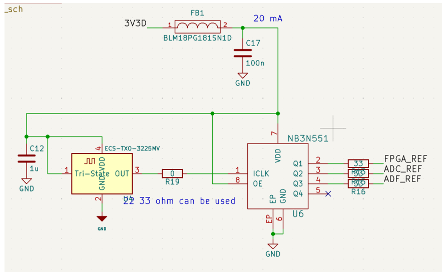


Sheet: /rx2/
File: rx.kicad_sch

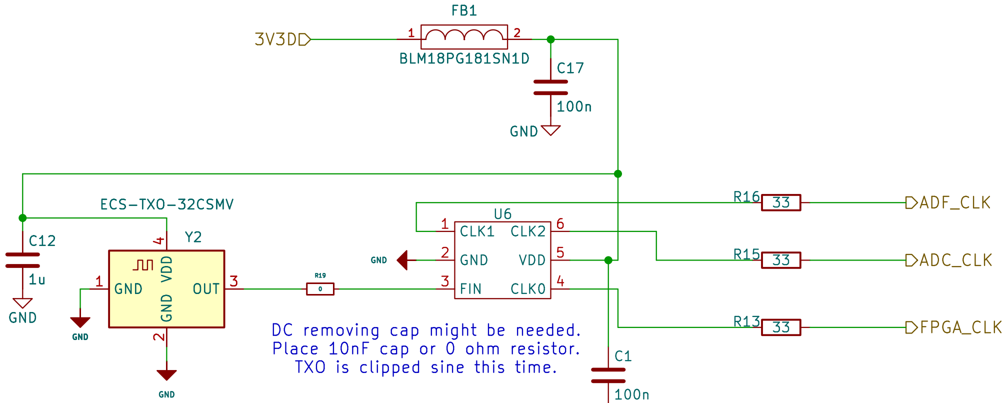
Title:

Size: A4 Date: **Rev:**

KiCad E.D.A. 9.0.7 Id: 10/12



This is the prev design:
NB3N551 is obs.
Also this txo is generating pure sin.
ADC might need cmos like input not sin.
That could be the reason of duty stabilizer.



DC removing cap might be needed.
Place 10nF cap or 0 ohm resistor.
TXO is clipped sine this time.

Sheet: /clock/
File: clock.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 11/12